

Santas Workshop

Every year, Kaggle (kaggle.com) post an optimization competition, always about helping Santa. In 2019, it was about planning workshop visits to Santa's workshop. The problem is a so-called Generalized Assignment problem.

Problem

- Minimize Santa Claus's visit costs (more money for presents !)

Sets

- Families: $f = 1, \dots, 5000$ or $f = 1, \dots, 1000$
- Days: $d = 1, \dots, 100$ or $d = 1, \dots, 20$

Parameters

- $FamilySize_f$: The number of members of family f
- $DayVisitCost_{f,d}$: Cost of assigning family f to day d

Decision variables

- $x_{f,d}$: 1 if family f visits the workshop on day d

Model

Objective:

- Minimize total Santa visit costs:

$$\sum_{f,d} DayVisitCost_{f,d} \cdot x_{f,d}$$

Constraints:

- Each family **HAS** to get a day:

$$\sum_d x_{f,d} = 1 \quad \forall f$$

- Each day, there has to be at least 125 persons in the workshop:

$$\sum_f FamilySize_f \cdot x_{f,d} \geq 125 \quad \forall d$$

- There are at most room for 300 persons in the workshop:

$$\sum_f FamilySize_f \cdot x_{f,d} \leq 300 \quad \forall d$$

The full model in Julia/JuMP, available with the name

`SantasWorkshop.jl`

from the book web-site, is given below:

```
*****
# Santas Workshop Assignment, Cannery example, "Mathematical Programming Modelling" (42112)
using JuMP
using HiGHS
using Printf
*****

*****
# Read data
# Data:
# F : No Families
# D : No Days
# FamilySize[f] : No of persons in a family
# DayVisitCost[f,d0] : Cost for the family visiting Santa that day
include("SantasWorkshopData_5000_100.jl")
#include("SantasWorkshopData_1000_20.jl")
println("No Families: ", F)
println("No Days: ", D)
*****

*****
```

```

# Model
santa = Model(HiGHS.Optimizer)

@variable(santa, x[f=1:F,d=1:D],Bin)

# minimize santa costs
@objective(santa, Min,
           sum( DayVisitCost[f,d]*x[f,d] for f=1:F,d=1:D) )

# force visit choice
@constraint(santa, [f=1:F],
           sum( x[f,d] for d=1:D) == 1)

# Day lower bound
@constraint(santa, [d=1:D],
           sum( FamilySize[f]*x[f,d] for f=1:F) >= 125)

# Day upper bound
@constraint(santa, [d=1:D],
           sum( FamilySize[f]*x[f,d] for f=1:F) <= 300)

#print(santa)
#####

#####
# solve
#set_silent(santa)
#set_optimizer_attributes(santa, "MIPGap" => 0.0)
println("Solving Model")
optimize!(santa)
#####

#####
# Report results
println("-----");
if termination_status(santa) == MOI.OPTIMAL
    println("RESULTS:")
    println("objective = $(objective_value(santa))\n")
else
    println(" No solution")
end
println("-----");
#####

```

Comments to the Julia/JuMP program:

- Notice: The data-set files are included into the program using the "include" statement, line 17 or line 18
- Notice: We comment out one of the two include statement, this makes it easy to switch between the data-sets, by simply changing which line is commented out.
- Notice: In the same way, the model statement which assigns the solver, is given in two different lines, line 26 and line 27, to make switching solver easy.